

Owen Lee

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EDUCATION

The University of Illinois Urbana-Champaign

Master of Computer Science

Urbana, IL

Expected May 2028

The Chinese University of Hong Kong, Shenzhen

Bachelor of Engineering, Computer Science and Engineering

Shenzhen, China

May 2026

Selected Coursework: Natural Language Processing, Machine Learning, Algorithms, Operating Systems, Computer Architecture

SKILLS

Programming Languages: Python, Java, C++, Javascript, HTML, SQL, D3.js, Plotly

AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Fine-tuning (LoRA/QLoRA), NLP, Generative AI

ML Frameworks: PyTorch, HuggingFace (Transformers, PEFT), NumPy, scikit-learn, Vector DBs (Weaviate, ChromaDB)

Technologies & Infra: Spring Boot, REST APIs, AWS (EC2, Lambda, Bedrock), Docker, Git, Linux

EXPERIENCE

Research Assistant (Advisor: Prof. Xiaoying Tang)

November 2025 – May 2026

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

- Engineered 1,040 step-grounded visual-text pairs for VLM training, contributing to ICML 2026 acceptance
- Designed CoT prompting templates aligning rationales with rendered intermediate states, improving training signal for visual reasoning benchmarks

Artificial Intelligence Engineering Intern

July 2025 – October 2025

ATG Nexus Sdn Bhd

Puchong, Malaysia

- Built hybrid keyword+vector retrieval (Weaviate + OpenAI embeddings) for a production legal search app
- Engineered bilingual triage report generator (Qwen3-Embedding-8B) and Gemini + Plotly Dash pipeline for automated insights, increasing analyst throughput by 50%

Research Assistant (Advisor: Prof. Benyou Wang)

January 2025 – May 2025

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

- Built multimodal crawler (1.2M audio-caption pairs) with CLAP-filtered pipeline, cutting hallucination by 15%
- Trained HTSAT+BERT on EC2-scaled S3 pipeline; achieved 44.3/57.8 R@1 (text→audio / audio→text), best among 5 baselines — contributing to ACL 2026 acceptance

Software Engineering Intern

May 2024 – August 2024

China Unicom

Zhejiang, China

- Built a Java and Spring Boot REST API for real-time weather dashboards, sustaining a 95% success rate at ~2,400 req/hr over 5 weeks in a high-scale production environment
- Implemented server health monitoring with 99% alert detection accuracy, validated against 4 weeks of alert logs

PROJECTS

MoE Routing vs. Dense Finetuning vs. Zero-Shot API | PyTorch, HuggingFace, QLoRA

Aug 2026

- Tested a router dispatching to 2B QLoRA specialists against a dense QLoRA 4B model and zero-shot Gemini, scoring 1,350 examples across 3 reasoning tasks with task-specific semantic verifiers
- Both finetuned approaches beat zero-shot API calls by an order of magnitude, and the dense 4B model outperformed the routed specialists by ~30% relative accuracy despite 100% router accuracy — showing task reasoning depth, not architecture, was the real bottleneck

Redline: LLM Red-Teaming Harness | FastAPI, Next.js, AWS Bedrock

Aug 2026 – Sep 2026

- Developed a red-teaming tool that tests whether LLMs can be tricked into unsafe or incorrect behavior, covering 8 attack types (prompt injection, jailbreaks, reasoning traps) across 4 AWS Bedrock models
- Auto-scored 800 model responses with an LLM-judge pipeline and found models blocked 88–100% of direct harmful requests but only 24–84% of prompt-injection attempts, showing refusal training generalizes poorly to indirect attacks

Retrieval-Grounded Text-to-SQL Chatbot | Python, MySQL, ChromaDB, SQLGlue

October 2025

- Architected schema-aware RAG pipeline (all-MiniLM + ChromaDB) for SQL generation grounded in live DB metadata
- Built an execution-based evaluation harness (500+ queries) to score generated SQL against live DB metadata, measuring 89% execution accuracy

PUBLICATIONS

Y. Huo, S. Zhang, K. Zeng, H. Liu, **O. Lee**, J. Chen, Y. Lu, Y. Guo, Y. Liang, X. Tang. *Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought*. [Paper](#). International Conference on Machine Learning (ICML), 2026.

S. Chen, X. Yuan, Z. Li, L. Zhao, **O. Lee**, Z. Su, Q. Sun, B. Wang. *FusionAudio-1.2M: Towards Fine-grained Audio Captioning with Multimodal Contextual Fusion*. [Paper](#). Association for Computational Linguistics (ACL), 2026.